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Contents

Linear algebra plays central role in the modern development of science and technology. It finds extensive applications in engineering, physics, computer science, economics, finance, cryptography, etc. It is arguably one of the most widely applied fields of mathematics. Due to the importance of linear algebra in various fields, in particular engineering, its basic theory has become indispensable.

This document presents lecture notes for Unit 1 of the course NMCI102 covering the following topics: matrices, vector spaces, linear transformation, eigenvalues and eigenvectors, diagonalization, and quadratic forms.

1 Matrices and system of linear equations

Matrices originated as a mathematical tool to represent and solve systems of linear equations efficiently, with their roots tracing back to ancient methods of organizing coefficients.

Definition 1. *A matrix is a rectangular array of scalars arranged in rows and columns. A general matrix of size (or order) $m \times n$ (where n is the number of rows and m is the number of columns) over a scalar field $\mathbb{F}$ is written as:*

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix},$$

where $a_{ij} \in \mathbb{F}$ represents the element of A in the i^{th} row and j^{th} column.

Depending on the structure of a matrix, some of the types of matrices are as follows:

- *Square matrix:* A matrix with the same number of rows and columns.
- *Diagonal matrix:* A square matrix where non-diagonal elements are zero.
- *Identity matrix:* A diagonal matrix where all diagonal elements are 1.
- *Zero matrix:* A matrix with all elements equal to 0.
- *Symmetric matrix:* A square matrix A such that $a_{ij} = a_{ji}$ for all i, j.
- *Skew-symmetric matrix:* A square matrix A such that $a_{ij} = -a_{ji}$ for all i, j.

Equality of matrices: Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are said to be equal if and only if they have the same size and the corresponding elements are equal, i.e., $a_{ij} = b_{ij}$ for all i, j.

1.1 Matrix operations

1.1.1 Addition of matrices and scalar multiplication

Matrix addition: Two matrices can be added if they have the same dimensions. The sum of two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ of the same order $m \times n$ is another matrix $C = [c_{ij}]$ whose elements are given by $c_{ij} = a_{ij} + b_{ij}$. We write C as $A + B$.

Scalar multiplication: Given a matrix $A = [a_{ij}]$ and a scalar α, the scalar multiplication of A with the scalar α is defined to be the matrix $\alpha A = [\alpha a_{ij}]$. In particular, we observe that $A + ((-1)A)$ is the zero matrix. So, we write $-A = (-1)A$.

1.1.2 Matrix multiplication

The multiplication (or product) of two matrices A and B is defined when their sizes are compatible; i.e., the number of columns in A must be equal to the number of rows in B. Under this compatibility condition, the multiplication of two matrices is defined as follows.

Definition 2. *Let A be a matrix of order $m \times n$ and B be a matrix of order $n \times p$. The multiplication of A and B is a matrix $C = AB$ is of the order $m \times p$, where the elements c_{ij} of C are given by*

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}, \quad \textit{for all } 1 \leq i \leq m, 1 \leq j \leq p.$$

The following observations are useful:

- This product is sometimes called the *row-column product* to emphasize the fact that it is a product involving the rows of A with the columns of B.
- The product of two matrices is not defined if their sizes are not compatible. For example, if A has size 2×5 and B has size 3×2 then their product AB is not defined. However, the product BA is defined. Why? What is the size of BA?
- Matrix multiplication distributes over addition, on both sides (provided, of course, that the

sizes of the matrices involved are compatible for the given operations):

$$(A+B)C = AC + BC \quad \text{and} \quad A(B+C) = AB + AC.$$

- Matrix multiplication is associative (again, under the compatibility condition):

$$(AB)C = A(BC).$$

- In particular, taking the n^{th} power of a square matrix is well-defined for every positive integer n.

- The transpose of the product of two matrices is the product of their transposes in reverse order:

$$(AB)^T = B^T A^T.$$

The matrix multiplication is illustrated in the following examples.

Example 3. *Let A and B be the following matrices:*

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}.$$

To compute $C = AB$, we find each element of C as:

$$C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix},$$

where

$$\begin{aligned} c_{11} &= (1)(5) + (2)(7) = 5 + 14 = 19, \\ c_{12} &= (1)(6) + (2)(8) = 6 + 16 = 22, \\ c_{21} &= (3)(5) + (4)(7) = 15 + 28 = 43, \\ c_{22} &= (3)(6) + (4)(8) = 18 + 32 = 50. \end{aligned}$$

Thus, the product is:

$$C = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}.$$

Example 4. *Let A and B be two 3×3 matrices given by*

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}.$$

The product $C = AB$ is computed as follows:

$$C = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix},$$

where elements of C are given by

$$c_{ij} = \sum_{k=1}^{3} a_{ik} b_{kj}, \qquad 1 \le i, j \le 3.$$

We thus have

$$\begin{aligned}
c_{11} &= (1)(9) + (2)(6) + (3)(3) = 9 + 12 + 9 = 30, \\
c_{12} &= (1)(8) + (2)(5) + (3)(2) = 8 + 10 + 6 = 24, \\
c_{13} &= (1)(7) + (2)(4) + (3)(1) = 7 + 8 + 3 = 18, \\
c_{21} &= (4)(9) + (5)(6) + (6)(3) = 36 + 30 + 18 = 84, \\
c_{22} &= (4)(8) + (5)(5) + (6)(2) = 32 + 25 + 12 = 69, \\
c_{23} &= (4)(7) + (5)(4) + (6)(1) = 28 + 20 + 6 = 54, \\
c_{31} &= (7)(9) + (8)(6) + (9)(3) = 63 + 48 + 27 = 138, \\
c_{32} &= (7)(8) + (8)(5) + (9)(2) = 56 + 40 + 18 = 114, \\
c_{33} &= (7)(7) + (8)(4) + (9)(1) = 49 + 32 + 9 = 90.
\end{aligned}$$

This gives

$$C = \begin{bmatrix} 30 & 24 & 18 \\ 84 & 69 & 54 \\ 138 & 114 & 90 \end{bmatrix}.$$

Example 5. *Let matrices A and B be given by*

$$A = \begin{bmatrix} 1 & 6 & -2 \\ 3 & 4 & 5 \\ 7 & 0 & 8 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -9 \\ 6 & 1 \\ 1 & -3 \end{bmatrix}.$$

Consider

$$C = AB = \begin{bmatrix} 1 & 6 & -2 \\ 3 & 4 & 5 \\ 7 & 0 & 8 \end{bmatrix} \begin{bmatrix} 2 & -9 \\ 6 & 1 \\ 1 & -3 \end{bmatrix}.$$

The first row of C is given by

$$\begin{aligned} c_{11} &= (1 \cdot 2) + (6 \cdot 6) + (-2 \cdot 1) = 2 + 36 - 2 = 36, \\ c_{12} &= (1 \cdot -9) + (6 \cdot 1) + (-2 \cdot -3) = -9 + 6 + 6 = 3. \end{aligned}$$

The second row of C is computed as

$$\begin{aligned} c_{21} &= (3 \cdot 2) + (4 \cdot 6) + (5 \cdot 1) = 6 + 24 + 5 = 35, \\ c_{22} &= (3 \cdot -9) + (4 \cdot 1) + (5 \cdot -3) = -27 + 4 - 15 = -38. \end{aligned}$$

Lastly, the third row of C is

$$\begin{aligned} c_{31} &= (7 \cdot 2) + (0 \cdot 6) + (8 \cdot 1) = 14 + 0 + 8 = 22, \\ c_{32} &= (7 \cdot -9) + (0 \cdot 1) + (8 \cdot -3) = -63 + 0 - 24 = -87. \end{aligned}$$

Therefore, the product AB is given by

$$C = \begin{bmatrix} 36 & 3 \\ 35 & -38 \\ 22 & -87 \end{bmatrix}.$$

1.1.3 Matrix transpose

If A is an $m \times n$ matrix then its transpose is the $n \times m$ matrix whose (i, j)-entry is equal to the (j, i)-entry of A. We denote the transpose of A by A^T. For example, the transpose of $A = \left[\begin{smallmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{smallmatrix}\right]$ is $A^T = \left[\begin{smallmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{smallmatrix}\right]$.

Recall that a square matrix A is said to be symmetric if it satisfies $A^T = A$, and it is called skew-symmetric if it satisfies $A^T = -A$.

1.1.4 Determinant of a matrix

Definition 6. *The determinant of an $n \times n$ matrix A, denoted by $\det(A)$ or $|A|$, is defined inductively as follows.*

- *For a 1×1 matrix $[a]$, it is defined as $\det[a] = a$.*
- *For $n \geq 2$, we compute the determinant via cofactor expansion: define $A^{(1,k)}$ to be the matrix obtained from A by deleting the 1st row and kth column. Then*

$$\det(A) = \sum_{k=1}^{n} (-1)^{k+1} a_{1,k} \det(A^{(1,k)}).$$

Below are explicit formulae for the determinants of 2×2 and 3×3 matrices.

- The determinant of a 2×2 matrix $\left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$ is given by $\det\left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right] = ad - bc$.
- The determinant of a 3×3 matrix $\left[\begin{smallmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{smallmatrix}\right]$ is given by

$$\det \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} = a_1 \det \begin{bmatrix} b_2 & b_3 \\ c_2 & c_3 \end{bmatrix} - a_2 \det \begin{bmatrix} b_1 & b_3 \\ c_1 & c_3 \end{bmatrix} + a_3 \det \begin{bmatrix} b_1 & b_2 \\ c_1 & c_2 \end{bmatrix}.$$

Example 7. *The determinant of the matrix* $\left[\begin{smallmatrix}1 & 2\\ 3 & 4\end{smallmatrix}\right]$ *is given by :*

$$\det\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix} = (1)(4) - (2)(3) = -2.$$

Example 8. *The determinant of* $\left[\begin{smallmatrix}1 & 2 & 4\\ -1 & 1 & 0\\ -2 & 1 & 3\end{smallmatrix}\right]$ *is given by*

$$\det\begin{bmatrix}1 & 2 & 4\\ -1 & 1 & 0\\ -2 & 1 & 3\end{bmatrix} = 1\det\begin{bmatrix}1 & 0\\ 1 & 3\end{bmatrix} - 2\det\begin{bmatrix}-1 & 0\\ -2 & 3\end{bmatrix} + 4\det\begin{bmatrix}-1 & 1\\ -2 & 1\end{bmatrix}.$$

$$= 1(3) - 2(-3) + 4(1) = 13.$$

1.1.5 Inverse of a matrix

Definition 9. *If* A *is an* $n \times n$ *matrix, then we say* A *is invertible (or nonsingular) if there exists another* $n \times n$ *matrix* B *such that*

$$AB = BA = I_n,$$

where I_n *is the* $n \times n$ *identity matrix. If such a* B *exists, it is unique. It is generally denoted by* A^{-1} *and is called the inverse of* A*. If* A *is not invertible, we call it non-invertible (or singular).*

For example, the matrix $A = \left[\begin{smallmatrix}2 & 5\\ 1 & 3\end{smallmatrix}\right]$ is invertible and its inverse is given by $A^{-1} = \left[\begin{smallmatrix}3 & -5\\ -1 & 2\end{smallmatrix}\right]$. One can easily check that

$$\begin{bmatrix}2 & 5\\ 1 & 3\end{bmatrix}\begin{bmatrix}3 & -5\\ -1 & 2\end{bmatrix}\begin{bmatrix}3 & -5\\ -1 & 2\end{bmatrix}\begin{bmatrix}2 & 5\\ 1 & 3\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}.$$

Not every matrix is invertible. An obvious example is the $n \times n$ zero matrix. Another example of non-invertible matrix is the $n \times n$ matrix with each element equal to 1 for $n \geq 2$. Indeed, we can see that

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a+c & b+d \\ a+c & b+d \end{bmatrix},$$

which is never equal to the identity matrix for any choice of a, b, c, d.

We enlist several properties of invertible matrices along with their proofs.

Proposition 10. *If a matrix is invertible, then its inverse is unique.*

Proof. Let A be an $n \times n$ invertible matrix. Suppose B_1 and B_2 both satisfy $AB_1 = I_n = B_1A$ and $AB_2 = I_n = B_2A$. Then $B_1 = B_1I_n = B_1(AB_2) = (B_1A)B_2 = I_nB_2 = B_2$ implying $B_1 = B_2$. □

Proposition 11. *If A and B are invertible $n \times n$ matrices with inverses A^{-1} and B^{-1}, then AB is also an invertible matrix with the inverse $B^{-1}A^{-1}$.*

Proof. We simply compute

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = A(I_n)A^{-1} = AA^{-1} = I_n.$$

Similarly, we also get $(B^{-1}A^{-1})(AB) = I_n$. □

Proposition 12. *Any square matrix with a row or column of all zeroes cannot be invertible.*

Proof. Suppose the $n \times n$ matrix A has all entries in its i-th row equal to zero. Then for any $n \times n$ matrix B, the product AB will have all entries in its i-th row equal to zero, so it cannot be the identity matrix.

Similarly, if the $n \times n$ matrix A has all entries in its i-th column equal to zero, then for any $n \times n$ matrix B, the product BA will have all entries in its i-th column equal to zero. □

Proposition 13. *A 2×2 matrix $\left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$ is invertible if and only if $ad - bc \neq 0$. In this case,* *the inverse is given by*

$$\frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Proof. This follows from solving the system of equations for e, f, g, h in terms of a, b, c, d that arises from comparing entries in the product

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

One obtains precisely the solution given above. If $ad = bc$, then the system is inconsistent and there is no solution; otherwise, there is exactly one solution as given. □

1.1.6 Elementary row operations for matrices

The following row operations on a matrix will be used frequently later.

- Interchanging two rows.
- Adding a scalar multiple of one row to another row.
- Multiplying a row by a nonzero constant.

The above row operations can be applied to any rectangular matrix to obtain a so-called *row-echelon form* of the matrix that we will see in the next section.

1.2 Row-echelon form and reduced row-echelon form

Here we shall briefly discuss the concepts of row-echelon and reduced row-echelon forms of rectangular matrices. These concepts will play important roles in solving systems of linear equations discussed in the next section.

1.2.1 Row-echelon form

Definition 14. *A rectangular matrix is said to be in row-echelon form if it satisfies the following two conditions:*

(i) *every row with a nonzero element is always above all the zero rows,*

(ii) *the first nonzero element in every row is always to the right of the first nonzero term in the row above it.*

The first nonzero element of a row is called the pivot element of that row.

The two conditions of Definition 14 can be restated in other words as follows:

(i') all rows without a pivot (the zero rows) are at the bottom,

(ii') any row's pivot, if it has one, lies to the right of the pivot of the row directly above it.

We will see in Section 1.4 that any rectangular matrix can be brought to a row-echelon form by performing elementary row operations on it. Below are some examples of matrices in row-echelon form whose pivots are boxed:

$$\begin{bmatrix} \boxed{1} & 2 & 3 & 4 & 5 \\ 0 & \boxed{-1} & 2 & 3 & 4 \\ 0 & 0 & \boxed{5} & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} \boxed{-7} & 2 & 3 & 4 & 5 \\ 0 & 0 & 0 & \boxed{8} & 0 \end{bmatrix}, \quad \begin{bmatrix} \boxed{10} & 2 & 3 & 4 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & \boxed{3} & 4 & 5 \\ 0 & 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Also, here are examples of matrices not in the row-echelon form. The matrix

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$

is not in the row-echelon form because the pivot in the second row is not strictly to the right of the pivot element in the row above it.

Another example of a matrix not in the row-echelon form is

$$\begin{bmatrix} 0 & 0 & 3 & 4 & 5 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix}.$$

It is because the pivot in the third row is not strictly to the right of the pivot in the second row.

Can you determine if the matrix

$$\begin{bmatrix} 0 & 0 & 3 & 4 & 5 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

is in the row-echelon form? Justify.

Remark: We will see in the next system that if the coefficient matrix of a system of linear equations is in a row-echelon form then it is easy to solve the system. It just requires backward substitution to obtain the solution.

1.2.2 Reduced row-echelon form

A matrix in row-echelon form can be further simplified into a form known as reduced row-echelon form through elementary row operations.

Example 15. *A rectangular matrix is said to be in reduced row-echelon form if it satisfies the following three conditions:*

(i) *it is in row-echelon form,*

(ii) *every pivot element is equal to* 1*,*

(iii) *the entries above every pivot element are all zero.*

Below are some examples of matrices in reduced row-echelon form with their pivots boxed:

$$\begin{bmatrix} \boxed{1} & 0 & 0 & 4 & 5 \\ 0 & \boxed{1} & 0 & 3 & 4 \\ 0 & 0 & \boxed{1} & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} \boxed{1} & 2 & 3 & 0 & 5 \\ 0 & 0 & 0 & \boxed{1} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} \boxed{1} & 2 & 0 & 4 & 0 \\ 0 & 0 & \boxed{1} & 3 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} \end{bmatrix}.$$

Also, here are some examples of matrices not in reduced row-echelon form. The matrix

$$\begin{bmatrix} 1 & 2 & 0 & 4 & 5 \\ 0 & 1 & 0 & 3 & 4 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$

is not in reduced row-echelon form because the entry above the second pivot element is not zero. The following matrix is also not in reduced row-echelon form.

$$\begin{bmatrix} 0 & 0 & 3 & 4 & 5 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Why?

As stated earlier, through elementary row-operations, every matrix can be brought to a reduced row-echelon form. It turns out that the reduced row-echelon form of a matrix is unique, as stated in the following theorem. We omit the proof.

Theorem 16. *Every matrix admits a unique reduced row-echelon form.*

Remark: We will later define the concept of *rank* of a matrix. For a matrix in row-echelon form, it is equivalently defined as the number of pivots in the matrix. For example, the rank of $\left[\begin{smallmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 0 & 1 \end{smallmatrix}\right]$ is 3, while the rank of $\left[\begin{smallmatrix} 1 & 2 & 3 & 0 & 5 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{smallmatrix}\right]$ is 2. More generally, the rank of any matrix A is equal to the rank of a matrix in row-echelon form obtained from A by applying elementary row operations.

1.3 System of linear equations

A system of linear equations consists of two or more linear equations involving the same set of variables. Such systems arise frequently in engineering, physics, computer science, and many other fields.

Definition 17. *A system of m equations in n variables is of the form:*

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2, \\
\vdots \quad &\quad \vdots \\
a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m,
\end{aligned}
$$

where $x_1, x_2, \ldots, x_n$ are the variables, a_{ij} are the coefficients, and $b_1, b_2, \ldots, b_m$ are the constants.

The above system of linear equations can be represented in the matrix form as follows:

$$A\mathbf{x} = \mathcal{B},$$

where A is an $m \times n$ matrix called the coefficient matrix, $\mathbf{x}$ is the column vector of variables, and $\mathcal{B}$ is the column vector of the constants given by:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \mathcal{B} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

The matrix $[A|\mathcal{B}]$ obtained by appending the constant vector $\mathcal{B}$ to the right of the coefficient matrix A is called the *augmented matrix* of the system. The vertical line in the augmented matrix is meant to separate the coefficients from the constants in the system. We will see later that working with the augmented matrix is a lot more convenient while solving a system of linear equations.

Some examples of the system of linear equations are:

$$
(1) \quad \begin{aligned} 2x + 3y &= 5, \\ 4x - y &= 11. \end{aligned}
$$

$$
(2) \quad \begin{aligned} x + 2y + z &= 6, \\ 3x - y + 4z &= 8, \\ 5x + y - 2z &= 4. \end{aligned}
$$

The traditional method for solving a system of linear equations (probably familiar from basic algebra) is by *elimination*: we solve the first equation for one variable, say x_1, in terms of the others. We then substitute the result into all other equations to obtain a reduced system involving one fewer variables. Eventually, the system is simplified to a scenario that implies a contradiction (e.g., $1 = 0$), a unique solution, or an infinite family of solutions.

Example 18. *Let us solve the following system of equations by the elimination method described above.*

$$\begin{aligned} x + y &= 7, \\ 2x - 2y &= -2. \end{aligned}$$

Step 1: Solve the first equation for x: $x = 7 - y$.

Step 2: Substitute this into the second equation to get:

$$2(7 - y) - 2y = -2 \quad \to \quad 14 - 4y = -2 \quad \to \quad y = 4.$$

Step 3: Substituting $y = 4$ into $x = 7 - y$ gives $x = 3$.

Thus, we get a unique solution to the system given by $(x, y) = (3, 4)$.

Another way to perform elimination is by adding or subtracting multiples of equations to eliminate variables, avoiding solving for individual variables explicitly.

Example 19. *Let us solve the following system of linear equations:*

$$\begin{aligned} x + y + 3z &= 4, && (i) \\ 2x + 3y - z &= 1, && (ii) \\ -x + 2y + 2z &= 1. && (iii) \end{aligned}$$

We first eliminate x by performing the following operations on the equations $(ii) \to (ii) - 2(i)$ and $(iii) \to (iii) + (i)$:

$$\begin{aligned} &x + y + 3z = 4, \\ &y - 7z = -7, \\ &3y + 5z = 5. \end{aligned}$$

Now, eliminate y by performing $(iii) \to (iii) - 3(ii)$:

$$
\begin{aligned}
x + y + 3z &= 4, && (i') \\
y - 7z &= -7, && (ii') \\
26z &= 26. && (iii')
\end{aligned}
$$

Solve equation (iii') to get $z = 1$. Substituting $z = 1$ into (ii') gives $y = 0$. Lastly, substitute $y = 0, z = 1$ into (i') to get $x = 1$. So, we get a unique solution $(x, y, z) = (1, 0, 1)$.

Observe that the above procedure of solving systems of linear equations requires dealing with only the coefficients of the variables and the constants. So, we only need to keep track of the coefficients, which we can do by putting them into an array. For example, the above system of linear equations given by $(i), (ii), (iii)$ can be written in simplified form using the array as

$$
\left[\begin{array}{ccc|c}
1 & 1 & 3 & 4 \\
2 & 3 & -1 & 1 \\
-1 & 2 & 2 & 1
\end{array}\right].
$$

Recall that the above matrix is called the augmented matrix for the given system of linear equations. We can then do operations on the entries in the array that correspond to manipulations of the associated system of equations. The elimination process consists of a combination of the three elementary row operations, which we recall below:

- Interchanging two rows.
- Adding a scalar multiple of one row to another row.
- Multiplying a row by a nonzero constant.

Each of these elementary row operations leaves unchanged the solutions to the associated system of linear equations. The idea of elimination is to apply these elementary row operations to the coefficient matrix until it is in a simple enough form that we can simply read off the solutions to the original system of equations. The above ideas lead to the Gaussian elimination process of solving systems of linear equations.

1.4 Gaussian elimination method for solving system of linear equations

1.4.1 The augmented matrix in a row-echelon form

If the augmented matrix of a system is in a row-echelon form, it is easy to read off the solutions to the corresponding system of linear equations by working from the bottom up. This is illustrated in the following example.

Example 20. *Consider the augmented matrix:*

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 2 & 4 \end{array}\right],$$

corresponding to the system:

$$\begin{aligned} x + y + 3z &= 4, \\ y - z &= 1, \\ 2z &= 4. \end{aligned}$$

The bottom equation immediately gives $z = 2$. Then the middle equation gives $y = 1 + z = 3$, and the top equation gives $x = 4 - y - 3z = -5$.

By using row operations on the augmented matrix, we can solve the associated system of equations, since, as we noted before, each of the elementary row operations does not change the solutions of the system.

By reducing the augmented matrix of a system to a row-echelon form, it becomes straightforward to conclude the *nature* of solutions. Given the augmented matrix in a row-echelon form, the following are the three possible scenarios.

Case 1: No solution. This happens precisely when two conditions are satisfied:

(i) each row contains a pivot element,

(ii) the pivot element in the last row appears in the last column.

Here is an example to illustrates this case. Consider the following augmented matrix in row-echelon form:

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & \boxed{4} \end{array}\right].$$

Note that the pivot element in the last row (boxed) appears in the last column. Its corresponding system is

$$\begin{aligned} x + y + 3z &= 4, \\ y - z &= 1, \\ 0 &= 4. \end{aligned}$$

The above system does not have any solution because the last equation is impossible to satisfy.

Case 2: Unique solution. This happens precisely when two conditions are satisfied:

(i) each row contains a pivot element,

(ii) the pivot element in the last row appears in the second-to-last column.

For example, consider the following augmented matrix in row-echelon form:

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & \boxed{2} & 4 \end{array}\right].$$

Observe that it satisfies the two conditions of the present case; i.e., each row has a pivot element and the pivot element of the last row (boxed) appears in the second-to-last column. Its corresponding system is

$$\begin{aligned} x + y + 3z &= 4, \\ y - z &= 1, \\ 2z &= 4. \end{aligned}$$

The unique solution is $z = 2$, $y = 3$, and $x = -5$, which we obtain by backward substitution..

Case 3: Infinitely many solutions. This happens precisely when the last column of the augmented matrix (in a row-echelon form) is zero. For example, consider the following augmented matrix in row-echelon form whose last row is zero:

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Its corresponding system is

$$\begin{aligned} x + y + 3z &= 4, \\ y - z &= 1, \\ 0 &= 0. \end{aligned}$$

The last equation is always true, so there are really only two relations between the three variables x, y, and z. By choosing arbitrary value of z, we can find values of the remaining two variables x and y in terms of z allowing for infinitely many solutions to exist for the system.

1.4.2 The Gaussian elimination process

The main idea behind the Gaussian elimination process to solve a system of linear equations is to bring the augmented matrix in a row-echelon form. Then depending on the nature of solutions, as discussed above, we either conclude there is not solution or solve the system through backward substitution.

The following is a step-by-step description of the Gaussian elimination process:

Step 1. **Convert the augmented matrix in a row-echelon form.** The augmented matrix can be reduced to a row-echelon form by performing elementary row operations on it. This is called forward elimination. If you go further with the process and convert the augmented matrix into its reduced row-echelon form then the following steps become significantly easier to execute; and this is called the Gauss–Jordan elimination process of solving system of linear equations.

Step 2. **Interpreting the nature of solutions:** If each row contains a pivot element and the pivot in the last row appears in the last column, then conclude that no solution exists. Else, proceed to the next step.

Step 3. **Calculating solutions explicitly:** If you are at this step, it means a solution exists. In this case, the variables corresponding to the pivot elements are given in terms of the remaining variables (also called free variables) through backward substitution. In particular, if there are n pivot elements for the system in n variables then the system has a unique solution. Else, the pivot variables are determined in terms of free variables, which implies infinitely many solutions.

We demonstrate the Gaussian elimination process through an example as follows.

Example 21. *Let us solve the system*

$$\begin{aligned} x + y + 3z &= 4, \\ 2x + 3y - z &= 1, \\ -x + 2y + 2z &= 1, \end{aligned}$$

using Gaussian elimination. The augmented matrix of the system is given by

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 4 \\ 2 & 3 & -1 & 1 \\ -1 & 2 & 2 & 1 \end{array}\right].$$

We now apply a series of elementary row operations to bring the augmented matrix in a row-echelon form as follows:

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 4 \\ 2 & 3 & -1 & 1 \\ -1 & 2 & 2 & 1 \end{array}\right]$$

Apply $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 + R_1$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 4 \\ 0 & 1 & -7 & -7 \\ 0 & 3 & 5 & 5 \end{array}\right]$$

Apply $R_3 \to R_3 - 3R_2$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 4 \\ 0 & 1 & -7 & -7 \\ 0 & 0 & 26 & 26 \end{array}\right]$$

We have now obtained a row-echelon form of the original augmented matrix. We can also see *that the system has a unique solution. Why? The solution can be easily obtained through back substitution, which is given by* $x = 1, y = 0, z = 1$.

1.5 Gauss–Jordan method for computing the inverse of a matrix

This section concerns with computation of the inverse of invertible matrices. The method described in the section is the Gauss–Jordan method for computing the inverse of a matrix. The main idea of this method is to convert the problem of finding the inverse of a matrix into solving several systems of linear equations simultaneously using the Gaussian elimination method.

Let A be an $n \times n$ invertible matrix. We know by definition that

$$AA^{-1} = I_n, \tag{1}$$

where recall that I_n is the identity matrix of size $n \times n$. Let us denote by $\mathbf{x}_1, \ldots, \mathbf{x}_n$ the (unknown) columns of A^{-1} and denote by $\mathbf{e}_1, \ldots, \mathbf{e}_n$ the columns of I_n. We can thus write (1) as

$$[A\mathbf{x}_1 \cdots A\mathbf{x}_n] = [\mathbf{e}_1 \cdots \mathbf{e}_n] .$$

The above matrix equation is equivalent to the following systems of linear equations:

$$A\mathbf{x}_i = \mathbf{e}_i, \quad 1 \leq i \leq n. \tag{2}$$

Therefore, finding the inverse of A boils down to solving the systems of linear equations (2) simultaneously. We apply to Gauss–Jordan elimination method (Gaussian elimination method with reduced row-echelon form of the augmented matrix) to solve the above systems of linear equations. This is known as the Gauss–Jordan method for computing the inverse of a matrix.

> Gauss–Jordan method for finding inverse $\equiv$ solving several systems of linear equations simultaneously

The following are the steps to perform Gauss–Jordan method to compute the inverse of a matrix:

Step 1. Consider the augmented matrix $[A|I_n]$.

Step 2. Reduce $[A|I_n]$ to its reduced row-echelon form:

(i) apply elementary row operations on $[A|I_n]$ to reduce it to a row-echelon form, say $[U|\star]$,

(ii) apply another set of elementary row operations on $[U|\star]$ to get reduced row-echelon form, which is given by $[I_n|A^{-1}]$.

Step 3. Read off A^{-1} from the reduced row-echelon form $[I_n|A^{-1}]$.

Gauss–Jordan method in a nutshell: $[A|I_n] \longrightarrow [U|\star] \longrightarrow [I_n|A^{-1}]$

We illustrate the Gauss–Jordan method in the following examples.

Example 22. *Let us find the inverse of* $\left[\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right]$ *using Gauss–Jordan method. Consider the augmented matrix* $[A|I_2]$*:*

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array}\right]$$

Apply $R_2 \to R_2 - \frac{1}{2}R_2$*:*

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} & 1 \end{array}\right]$$

Apply $R_1 \to \frac{1}{2}R_1, R_2 \to 2R_2$*:*

$$\left[\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & -1 & 2 \end{array}\right]$$

Apply $R_1 \to R_1 - \frac{1}{2}R_2$*:*

$$\left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \end{array}\right]$$

The inverse of $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ *is thus given by* $\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$.

Example 23. *Consider the matrix*

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Let us find its inverse using the Gauss–Jordan method. Consider the augmented matrix $[A|I_3]$*:*

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

Apply on the above matrix $R_2 \to R_2 - \frac{1}{2}R_1$*:*

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & \frac{3}{2} & 1 & -\frac{1}{2} & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

Apply $R_3 \to R_3 - \frac{2}{3}R_2$*:*

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & \frac{3}{2} & 1 & -\frac{1}{2} & 1 & 0 \\ 0 & 0 & \frac{4}{3} & \frac{1}{3} & -\frac{2}{3} & 1 \end{array}\right]$$

Apply $R_1 \to \frac{1}{2}R_1,\ R_2 \to \frac{2}{3}R_2,\ R_3 \to \frac{3}{4}R_3$*:*

$$\left[\begin{array}{ccc|ccc} 1 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 1 & \frac{2}{3} & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 1 & \frac{1}{4} & -\frac{1}{2} & \frac{3}{4} \end{array}\right]$$

Apply $R_1 \to R_1 - \frac{1}{2}R_2$*:*

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{2}{3} & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 1 & \frac{1}{4} & -\frac{1}{2} & \frac{3}{4} \end{array}\right]$$

Apply $R_1 \to R_1 + \frac{1}{3}R_3,\ R_2 \to R_2 - \frac{2}{3}R_3$*:*

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{3}{4} & -\frac{1}{2} & \frac{1}{4} \\ 0 & 1 & 0 & -\frac{1}{2} & 1 & -\frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{4} & -\frac{1}{2} & \frac{3}{4} \end{array}\right]$$

The inverse of A *is thus given by*

$$A^{-1} = \begin{bmatrix} \frac{3}{4} & -\frac{1}{2} & \frac{1}{4} \\ -\frac{1}{2} & 1 & -\frac{1}{2} \\ \frac{1}{4} & -\frac{1}{2} & \frac{3}{4} \end{bmatrix}.$$